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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Integrated Hotel Booking And Guest Experience Management System

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA4305-INTERNET PROGRAMMING

to the award of the degree of

BACHELOR OF ENGINEERING

IN

**COMPUTER SCIENCE AND ENGINEERING
(CSE-AI)**

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Integrated Hotel Booking And Guest Experience Management System**” has been carried out by **K.Mrudhula(192372290)** under the supervision of **DR D Shudhagaran** and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech Computer Science and Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **K. Mrudhula(192372290)** of the **CSE AI**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Integrated Hotel Booking And Guest Experience Management System**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

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INDIVIDUAL CONTRIBUTION STATEMENT

In accordance with capstone evaluation guidelines, the detailed individual responsibilities and specific architectural contributions of each group member are delineated below:

Student Name / Reg No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. (%)
Mrudhula.k (192372290)	User Authentication & Student Profile Management	Designed registration, login, authentication, profile creation, and profile-management functionality	Tested login validation, registration forms, profile updates, and invalid-input handling	Abstract, Chapter 1, and requirements documentation	33.3%
P.Harika (192311243)	Skill Management & Skill Assessment	Developed technical/professional skill management, assessment questions, answer evaluation, score calculation, and skill-level classification	Tested assessment functionality, score calculation, different score ranges, and skill-level results	Chapter 2, assessment methodology, and related documentation	33.3%
N.Vishnu Sai Sree (192372039)	Learning Resources, Progress Tracking & System Integration	Developed learning-resource management, assessment-history, progress tracking, dashboard integration, and overall module integration	Conducted integration testing, progress verification, resource testing, debugging, and system-level testing	Chapters 3, 4, 5, and Appendices	33.4%
Total	Complete Student Skill Development Management System	Full End-to-End Web Application Development	Functional, Integration & System Testing	Complete Technical Documentation	100%

ABSTRACT

The Student Skill Development Management System is a web-based application designed to provide a centralized platform for managing, assessing, developing, and monitoring the technical and professional skills of students. In the present educational environment, students acquire skills through academic courses, projects, workshops, certifications, internships, and online learning platforms. However, maintaining skill information and monitoring individual development across different sources can be difficult. The proposed system addresses this issue by bringing important skill-development activities together within a single web-based platform.

The system allows students to register and log in, manage their personal profiles, add and update technical and professional skills, participate in skill assessments, view assessment results, access suitable learning resources, and monitor their progress. The assessment module evaluates student responses and calculates scores to determine the corresponding proficiency level. Assessment history can be maintained to compare previous and current performance, helping students understand their strengths and identify areas requiring improvement. The system also provides administrative functionality for managing student information, skills, assessment questions, and learning resources.

The application demonstrates the practical implementation of Internet Programming concepts, including HTML for web-page structure, CSS for interface design and styling, JavaScript for dynamic interaction and validation, and appropriate server-side and database technologies for processing and storing information. The system follows a modular architecture consisting of user authentication, student profile and skill management, skill assessment, learning resources, progress tracking, and administrator management modules. The application is tested using functional test cases to verify registration, authentication, skill management, assessment, result generation, resource access, and progress tracking.

The proposed system provides a systematic approach to student skill development by combining skill identification, assessment, learning support, and progress monitoring in one platform. It can reduce manual management, improve accessibility of student skill information, support self-evaluation, and provide useful information for faculty and administrators. Future enhancements can include AI-based personalized recommendations, advanced performance analytics, certification tracking, mobile application support, and integration with external learning platforms.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Abbr.	Description	Standard / Unit
IP	Internet Programming	Course / Domain
HTML	HyperText Markup Language	Web Standard
CSS	Cascading Style Sheets	Web Standard
JS	JavaScript	Programming Language
DOM	Document Object Model	Web Programming
UI	User Interface	Software
UX	User Experience	Software Design
HTTP	HyperText Transfer Protocol	Web Protocol
HTTPS	HyperText Transfer Protocol Secure	Secure Web Protocol
URL	Uniform Resource Locator	Web Address
API	Application Programming Interface	Software Interface
JSON	JavaScript Object Notation	Data Format
SQL	Structured Query Language	Database Language
DB	Database	Data Storage
DBMS	Database Management System	Software
CRUD	Create, Read, Update, Delete	Database Operations
REST	Representational State Transfer	Web Architecture
AJAX	Asynchronous JavaScript and XML	Web Technology
PHP	Hypertext Preprocessor	Server-Side Language
IDE	Integrated Development Environment	Software Tool
IDE	Integrated Development Environment	Development Tool
MVC	Model-View-Controller	Software Architecture
JWT	JSON Web Token	Authentication / Security
Auth	Authentication	Security
RBAC	Role-Based Access Control	Security
OTP	One-Time Password	Authentication
ERD	Entity-Relationship Diagram	Database Design
LMS	Learning Management System	Educational System
SDG	Sustainable Development Goals	UN Framework
OWASP	Open Worldwide Application Security Project	Web Security
WCAG	Web Content Accessibility Guidelines	Accessibility Standard
FR	Functional Requirement	Requirement Specification
NFR	Non-Functional Requirement	Requirement Specification
API	Application Programming Interface	Software

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

In the present digital and competitive educational environment, students are expected to develop a combination of academic knowledge, technical skills, and professional competencies. Skills such as programming, web development, database management, communication, teamwork, problem-solving, and leadership have become important for higher education and career development. Students acquire these skills through academic courses, projects, internships, workshops, certifications, and online learning platforms.

However, student skill information is often maintained through separate records, certificates, documents, or individual learning platforms. This makes it difficult for students to maintain an updated skill profile, evaluate their current competency level, identify skill gaps, and monitor their improvement over time. Faculty members and administrators may also face difficulties in systematically managing and reviewing the skill-development information of a large number of students.

The Student Skill Development Management System is proposed as a web-based application to provide a centralized platform for managing student skills and development activities. The system allows students to create profiles, maintain their technical and professional skills, participate in skill assessments, view their results, access relevant learning resources, and monitor their progress.

The proposed system applies Internet Programming concepts to provide an interactive and accessible platform. Web technologies such as HTML, CSS, and JavaScript can be used for developing the user interface and interactive features, while server-side programming and database technologies can be used for processing and storing student, skill, and assessment information.

1.2 Need for the Project

Students currently use different platforms and methods to learn and record their skills. Academic records mainly represent course performance, while certificates and project documents may represent specific achievements. However, these sources do not always provide a centralized view of a student's current technical and professional capabilities.

Manual skill tracking also makes continuous monitoring difficult. Students may know that they have learned a particular skill but may not have a systematic method to assess their proficiency or compare their performance over time. Similarly, faculty members may need to depend on separate records when reviewing student development.

Therefore, there is a need for an integrated web-based system that combines student profile management, skill management, skill assessment, learning resources, and progress tracking within a single platform.

The Student Skill Development Management System addresses this need by providing:

- Centralized student skill-profile management.
- Online skill assessment and score calculation.
- Identification of student skill levels.
- Access to relevant learning resources.
- Monitoring of skill-development progress.
- Administrative management of students, skills, assessments, and resources.

1.3 Problem Statement

The engineering problem addressed by this project is to design and implement a web-based system capable of managing student skill information and supporting continuous skill development. The system must collect student information, maintain technical and professional skills, conduct skill assessments, calculate assessment results, identify proficiency levels, provide appropriate learning resources, and maintain progress information.

The system should reduce the difficulty associated with maintaining separate skill records and provide students with a centralized platform for understanding and improving their competencies. It should also provide administrators or faculty members with suitable facilities for managing student-related skill information and assessment resources.

1.4 Project Aim

The primary aim of this project is to develop a **Student Skill Development Management System**, a web-based application that integrates student profile management, skill assessment, learning support, and progress tracking into a single platform.

The system aims to provide students with an organized method for identifying their current skill levels, accessing appropriate learning resources, and monitoring their improvement over time while providing administrative facilities for managing the system's skill and assessment information.

1.5 Project Objectives

The project is developed with the following objectives:

- **Objective 1:** To develop a web-based platform for centralized student skill management.
- **Objective 2:** To provide secure student registration and login functionality.
- **Objective 3:** To allow students to create and maintain their technical and professional skill profiles.
- **Objective 4:** To design and implement an online skill assessment mechanism for evaluating student competency.

- **Objective 5:** To automatically calculate assessment scores and determine the corresponding skill level.
- **Objective 6:** To provide suitable learning resources based on the student's skills and assessment results.
- **Objective 7:** To maintain assessment history and track student skill-development progress.
- **Objective 8:** To provide administrative functionality for managing students, skills, assessments, and learning resources.
- **Objective 9:** To apply relevant Internet Programming concepts in the development of a practical web application.
- **Objective 10:** To reduce manual effort involved in maintaining and monitoring student skill-development information.

1.6 Scope

The scope of the Student Skill Development Management System includes the development of a web-based application for managing student skills and monitoring their development.

Included Scope

- Student registration and login.
- Student profile management.
- Technical and professional skill management.
- Skill-level management.
- Online skill assessments.
- Automatic assessment-score calculation.
- Skill-level identification.
- Learning-resource management.
- Assessment-history management.
- Student progress tracking.
- Student performance display.
- Administrator management of system information.

Excluded Scope

The initial system does not focus on complete academic management such as attendance, examination management, fee management, or university-wide student administration.

Advanced features such as AI-based skill prediction, automated job placement, and integration with external recruitment platforms are considered possible future enhancements.

Assumptions

- Students provide valid personal and skill information.
- Assessment questions are relevant to the selected skills.
- Students have access to a compatible web browser.
- The application database is available during system operation.
- Learning resources stored in the system are relevant to the corresponding skills.

Boundary Conditions

- Only registered users can access personalized student functions.
- Assessment scores are calculated according to predefined evaluation rules.
- Progress tracking depends on the availability of previous assessment records.
- Learning recommendations are limited to the resources available in the system.
- Administrative functions are accessible only to authorized users.

1.7 Expected Engineering Outcome

The expected outcome of the project is a functional web-based **Student Skill Development Management System** that provides an integrated workflow for student skill development.

The expected execution flow is:

Student Registration → Login → Profile Management → Skill Management → Skill Assessment → Score Calculation → Skill-Level Identification → Learning Resources → Progress Tracking

The system is expected to provide a centralized and organized approach to student skill management, reduce manual record maintenance, support self-assessment, and help students monitor their skill-development progress. It also demonstrates the practical application of Internet Programming concepts in developing a complete web-based management system.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

To establish a suitable foundation for the **Student Skill Development Management System**, a structured review of existing student skill-management platforms, online assessment systems, learning management systems, and web-based educational applications was considered. The review focused on systems that support student profile management, skill identification, online assessments, learning resources, performance tracking, and administrative management.

The reviewed systems were evaluated across four major analytical dimensions: **(1) student profile and skill management, (2) assessment and skill-level identification, (3) learning-resource and recommendation support, and (4) progress monitoring and usability**. These dimensions were selected to identify the capabilities and limitations of existing approaches and to determine the requirements for an integrated web-based solution.

2.2 Review of Existing Work

Traditional Manual Skill Management Systems

Traditional student skill management is often performed through paper records, spreadsheets, classroom activities, certificates, and separate academic records. Students and faculty may maintain skill information independently, making it difficult to obtain a centralized and continuously updated view of a student's technical and professional abilities.

Learning Management Systems

Learning Management Systems (LMS) provide digital environments for delivering educational content, assignments, assessments, and student activities. Such platforms improve accessibility and allow students to interact with learning materials through web interfaces.

Online Skill Assessment Systems

Online assessment platforms provide structured questions, automated evaluation, score calculation, and immediate feedback. Objective questions can be evaluated automatically, allowing students to identify their performance in a particular subject or skill area.

However, many assessment-oriented systems focus mainly on conducting tests and displaying marks. They may not maintain a complete history of a student's skills, connect assessment results with learning resources, or provide an integrated mechanism for tracking skill-development progress.

Student Profile and Skill Management Systems

Modern educational web applications provide digital student profiles containing personal, academic, and extracurricular information. Some systems also allow users to record technical skills, certifications, projects, and professional competencies.

Personalized Learning and Recommendation Systems

Recommendation-based educational systems attempt to provide learning materials according to student interests, previous performance, or selected subjects. Such systems can help students identify suitable courses, tutorials, videos, and other educational resources.

2.3 Comparative Analysis

Table 2.1 presents a comparative evaluation of existing approaches against the proposed **Student Skill Development Management System**.

Platform / System	Skill Profile Management	Online Assessment	Learning Resource Support	Progress Tracking	Integrated Usability
Manual Records / Spreadsheets	Moderate (Manual entry)	Low (Separate process)	Low	Low	Low
Learning Management Systems	Moderate (Course-oriented)	High	High	Moderate	High
Online Assessment Systems	Low	High	Low	Moderate	Moderate
Student Portfolio Systems	High	Low	Moderate	Low	High
Recommendation-Based Learning Systems	Moderate	Moderate	High	High	High
Student Skill Development Management System (Proposed)	High (Centralized skill profile)	High (Automated evaluation)	High (Skill-based resources)	High (Assessment history and progress)	High (Integrated web dashboard)

2.4 Research / Engineering Gap

The review of existing approaches identifies several important engineering gaps:

- **Siloed Student Skill Information:** Existing educational platforms often maintain academic, assessment, portfolio, and skill information separately. This makes it difficult for students to obtain a single consolidated view of their skill development.
- **Disconnect Between Assessment and Skill Profile:** Many online assessment systems calculate marks but do not automatically use those results to update or identify the student's skill level.

- **Limited Skill-Based Learning Support:** Conventional academic systems provide general learning materials, but they may not connect resources directly with the student's identified skill requirements or assessment performance.
- **Insufficient Continuous Progress Tracking:** Manual records and basic assessment systems provide limited support for comparing previous and current performance to understand skill improvement.
- **Lack of an Integrated Web-Based Solution:** Existing approaches generally concentrate on one specific function such as course delivery, assessment, portfolio creation, or resource recommendation. There is a need for a unified system combining these functions within a single web application.

2.5 Proposed Contribution

The proposed **Student Skill Development Management System** addresses these identified gaps through an integrated web-based approach:

- **Unified Student Skill Profile:** Combines student information, technical skills, professional skills, and skill levels within a centralized profile.
- **Integrated Skill Assessment:** Provides online assessments for selected skills and automatically calculates scores based on student responses.
- **Skill-Level Identification:** Uses assessment performance and predefined criteria to classify the student's proficiency level, such as **Beginner, Intermediate, or Advanced**.
- **Skill-Based Learning Support:** Connects identified skills with relevant learning resources to help students improve areas where additional development is required.
- **Continuous Progress Monitoring:** Maintains assessment history and displays changes in performance so that students can monitor their development over time.
- **Administrative Management:** Provides authorized administrators with facilities to manage student information, skills, assessment questions, and learning resources.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

To establish clear engineering benchmarks, the system requirements for the **Student Skill Development Management System** were identified based on the needs of students, administrators, and the overall educational environment. These requirements were translated into functional and non-functional specifications covering user management, skill assessment, learning resources, progress tracking, and system security.

Stakeholder	Primary Need / Goal	Derived Engineering Requirement
Student	Maintain and improve technical and professional skills	Centralized student profile, skill management, assessment, learning resources, and progress tracking
Faculty / Mentor	Understand student skill levels and development areas	Access to assessment results, skill levels, and progress information
System Administrator	Maintain student, skill, assessment, and resource information	Role-based administration, database management, and secure access
Institution	Provide a structured digital skill-development platform	Reliable web-based system with organized student skill records and reports

Functional and Non-Functional Requirements

Requirement ID	Category	Description	Measurable Target
FR-01	Functional	Student registration and login	Successful authentication for registered users
FR-02	Functional	Student profile management	Students can create and update profile information
FR-03	Functional	Technical and professional skill management	Students can add, update, and remove skills
FR-04	Functional	Online skill assessment	Students can attempt predefined skill-based assessments

FR-05	Functional	Automatic score calculation	Score generated immediately after assessment submission
FR-06	Functional	Skill-level identification	Classify performance into predefined levels
FR-07	Functional	Learning resource management	Display resources associated with selected skills
FR-08	Functional	Progress tracking	Maintain assessment history and display performance changes
FR-09	Functional	Administrator management	Authorized administrator can manage system data
NFR-01	Non-Functional	Usability	Provide simple and responsive web interface
NFR-02	Non-Functional	Performance	Common user operations should respond without noticeable delay
NFR-03	Non-Functional	Reliability	Maintain consistent database records during normal operation
NFR-04	Non-Functional	Security	Protect user credentials and restrict unauthorized access

3.2 Constraints & Applicable Standards

The development of the Student Skill Development Management System is subject to technological, usability, security, and resource constraints. Since the application is designed as an Internet Programming project, standard web-development practices are considered to improve reliability, accessibility, and security.

Standard / Guideline	Issuing Body	Scope & Requirements	Implementation in Proposed System
HTML5	W3C	Structured web-page development	Semantic structure for system pages
CSS3	W3C	Web presentation and responsive styling	Layout, styling, and responsive interface
JavaScript / ECMAScript	ECMA International	Client-side programming	Form validation and interactive functionality
HTTP/HTTPS	IETF	Web communication	Communication between browser and server

WCAG	W3C	Web accessibility	Readable content, appropriate labels, and accessible interface
OWASP Web Security Guidelines	OWASP	Secure web application development	Input validation, authentication, and protection against common attacks

Major Constraints

- The system depends on the availability of a web browser and server environment.
- Skill levels are determined using predefined assessment rules rather than advanced artificial intelligence.
- Learning recommendations are limited to the resources stored in the system.
- Assessment accuracy depends on the quality and relevance of the questions provided.
- The system is primarily designed for student skill development and does not replace complete academic management systems.
- Database availability is required for persistent storage of profiles, skills, assessments, and progress records.

3.3 Design Specifications

Table 3.3 defines the major design parameters considered for the proposed system.

Design Parameter	Specification Target	Engineering Unit	Priority
Student registration	Complete registration with required details	User account	Critical
Authentication	Valid username/email and password	Login request	Critical
Skill categories	Technical and professional skills	Skill records	High
Assessment questions	Predefined skill-based questions	Questions/assessment	High
Score calculation	Automatic calculation after submission	Percentage	Critical
Skill classification	Beginner / Intermediate / Advanced	Skill level	High
Learning resources	Resources mapped to skills	Resource records	High
Progress history	Store previous assessment results	Assessment records	High

Database integrity	Consistent storage and retrieval	Database records	Critical
Interface responsiveness	Usable on common screen sizes	Web interface	Medium
Access control	Student and administrator roles	User role	Critical

3.4 Alternative Solutions & Evaluation

During system design, three alternative approaches were considered for implementing the Student Skill Development Management System.

Alternative 1: Manual / Spreadsheet-Based System

This approach stores student profiles, skills, and assessment results using spreadsheets or manually maintained documents. It is inexpensive and simple to start with, but it provides limited automation, weak data consistency, and no integrated online assessment or progress-management workflow.

Alternative 2: Separate Web Applications

In this approach, different applications are used for student profiles, online assessments, learning resources, and progress monitoring. Each individual application can perform its task effectively, but students must use multiple systems and information becomes distributed across different platforms.

Alternative 3: Integrated Web-Based Skill Management System

The proposed approach combines authentication, profile management, skill management, assessment, score calculation, learning resources, and progress tracking into one application. This provides centralized data management and a consistent user experience while remaining suitable for an academic Internet Programming project.

Evaluation Criterion	Weight	Manual Spreadsheet	Separate Web Applications	Integrated Proposed System
Skill Management	25%	2 / 5	4 / 5	5 / 5
Assessment Automation	25%	1 / 5	4 / 5	5 / 5
Progress Tracking	20%	2 / 5	4 / 5	5 / 5
Usability	15%	2 / 5	3 / 5	5 / 5
Maintenance & Integration	15%	3 / 5	2 / 5	4 / 5

Justification: The integrated web-based approach provides the best balance between functionality, usability, maintainability, and implementation feasibility. It eliminates the

fragmentation associated with manual records and separate applications while providing a centralized platform for student skill development.

3.5 Selected Design & Detailed Engineering Architecture

The proposed system is organized into six functional modules operating as an integrated web-based pipeline:

Module 1: User Authentication and Account Management

This module manages student registration, login, logout, and user access. Authentication ensures that only registered users can access protected student functions, while administrator access is restricted according to the assigned role.

Module 2: Student Profile and Skill Management

This module maintains student information and provides facilities for adding and updating technical and professional skills. The student's selected skills form the foundation for subsequent assessment and learning-resource activities.

Module 3: Skill Assessment

The assessment module presents predefined questions related to selected skills. Student responses are evaluated automatically, and the system calculates the total score and percentage.

Module 4: Learning Resource and Recommendation

Based on the selected skills and assessment performance, the system displays appropriate learning materials such as tutorials, courses, documentation, or practice resources. This helps students focus on areas requiring improvement.

Module 5: Progress Tracking and Performance Analysis

This module stores previous assessment results and presents the student's performance history. Comparison of previous and current scores allows students to observe their skill-development progress.

Module 6: Administrator Management

The administrator manages student records, skill categories, assessment questions, and learning resources. Administrative functions ensure that the information used throughout the system remains organized and updated.

Overall System Flow

Student Registration → Login → Profile Management → Skill Selection → Skill Assessment → Score Calculation → Skill-Level Identification → Learning Resources → Progress Tracking

3.6 Trade-off Analysis

Engineering design requires selecting suitable technologies and approaches while considering simplicity, performance, maintainability, and functionality.

Design Decision	Alternative Option	Evaluated Advantage	Evaluated Penalty	Resolution & Justification
Integrated Web Application	Separate applications	Centralized information and simpler user workflow	Requires integration of multiple modules	Selected integrated architecture for unified management
Rule-Based Skill Classification	Machine Learning Classification	Simple, explainable, and easy to implement	Less adaptive than ML	Selected rule-based approach because assessment thresholds are sufficient
Relational Database	NoSQL Database	Strong relationships and structured records	Requires predefined schema	Selected relational database for students, skills, assessments, and resources
Client-Side Validation	Server-only validation	Immediate feedback to users	Cannot replace server-side security	Selected as an additional validation layer
Responsive Web Interface	Fixed-size interface	Works across different devices	Requires additional CSS design	Selected responsive design for improved accessibility

3.7 Sustainability, Ethics, Safety, Risk & Security

The Student Skill Development Management System considers sustainability, ethical use of student information, system safety, and security throughout its design.

Domain	Consideration	Evidence Evaluated	Impact on System Design
Sustainability	Reduction of paper-based records	Digital storage reduces dependency on physical documents	Student skill and assessment records are maintained electronically
Ethics	Responsible use of student information	Student profiles and performance data are personal information	Access is restricted to authorized users
Fairness	Consistent skill evaluation	Manual evaluation can vary between evaluators	Predefined assessment rules provide consistent scoring

Safety	Prevention of incorrect system actions	Invalid inputs may cause errors or inconsistent records	Form validation and controlled operations are applied
Security	Protection against unauthorized access	Student records and credentials require protection	Authentication, authorization, input validation, and secure database practices are used

Risk Register and Mitigation Strategy

Identified Risk	Likelihood	Severity	Risk Level	Mitigation Strategy	Residual Risk
Unauthorized account access	Medium	High	High	Authentication and role-based access control	Low
Incorrect assessment questions	Medium	Medium	Medium	Review and validate questions before deployment	Low
Incorrect skill classification	Medium	Medium	Medium	Clearly defined score thresholds and testing	Low
Database failure / data loss	Low	High	Medium	Regular database backup and validation	Low
Invalid user input	High	Medium	High	Client- and server-side validation	Low
Unauthorized modification of records	Low	High	Medium	Administrator authorization and access restrictions	Low
Server unavailability	Medium	Medium	Medium	Proper server configuration and error handling	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The **Student Skill Development Management System** was developed using an iterative software development approach, where the system was divided into smaller functional modules. Each module was designed, implemented, tested, and integrated progressively. This approach helped in identifying errors at an early stage and ensured that the authentication, student profile, skill management, assessment, learning resources, and progress-tracking components worked together correctly.

The development process focused on separating the user interface, application logic, and database operations. This separation improves maintainability and allows individual modules to be modified or tested without affecting the complete system.

Resource Category	Resource Specification	Purpose & Architectural Role
Development Hardware	Computer/Laptop with adequate RAM and storage	Application development, testing, and execution
Operating System	Windows / Linux	Development and execution environment
Web Browser	Google Chrome / Microsoft Edge	Running and testing the web application
Database	Relational database	Storage of student, skill, assessment, and resource data
Testing Data	Sample student profiles and assessment questions	Functional and system testing
Version Control	Git / GitHub	Source-code management and project submission

4.2 Modern Software Tools and Frameworks

Table 4.2 presents the major technologies and tools used for developing the proposed system.

Tool / Technology	Type / Layer	Functional Scope	Project Phase
HTML5	Frontend	Structure of web pages	Development
CSS3	Frontend	Styling, layout, and responsive design	Development
JavaScript	Frontend Logic	Dynamic interaction and validation	Development

Backend Technology	Server Layer	Application logic and request processing	Development
MySQL / Relational Database	Database	Student, skill, assessment, and resource storage	Data Management
Git	Version Control	Source-code version management	Development
GitHub	Repository	Code storage and project collaboration	Submission
Web Browser	Testing	Application execution and interface testing	Verification
VS Code	Development Environment	Source-code development and debugging	Development

4.3 Test Plan & Verification

A comprehensive testing process was performed to verify the functionality and reliability of the system. **Unit testing** was used to validate individual modules, **integration testing** was used to verify communication between modules, and **system testing** was performed to evaluate the complete application workflow.

Test Module	Test Method & Procedure	Acceptance Criterion	Outcome
User Registration	Enter valid student details and submit registration form	Account should be created successfully	Pass
User Login	Enter valid and invalid credentials	Valid user should login; invalid user should be rejected	Pass
Profile Management	Add and update student information	Updated information should be stored correctly	Pass
Skill Management	Add, edit, and delete technical/professional skills	Skill records should be updated correctly	Pass
Skill Assessment	Attempt questions and submit answers	System should calculate score automatically	Pass
Skill-Level Identification	Test different score ranges	Correct proficiency level should be displayed	Pass
Learning Resources	Select a skill and view associated resources	Relevant resources should be displayed	Pass

Progress Tracking	Complete multiple assessments	Previous results should be stored and displayed	Pass
Administrator Module	Login as administrator and manage records	Only authorized administrator should access management functions	Pass
Input Validation	Enter incomplete/invalid data	System should display appropriate validation message	Pass

Requirement Verification and Traceability Matrix

Req ID	Specification Description	Observed Result	Compliance Status
FR-01	Student Registration and Login	Registration and authentication completed successfully	Fully Compliant
FR-02	Student Profile Management	Profile information successfully created and updated	Fully Compliant
FR-03	Skill Management	Technical and professional skills stored successfully	Fully Compliant
FR-04	Online Skill Assessment	Assessment questions displayed and evaluated	Fully Compliant
FR-05	Automatic Score Calculation	Score generated after submission	Fully Compliant
FR-06	Skill-Level Identification	Correct level generated from assessment score	Fully Compliant
FR-07	Learning Resource Support	Skill-related resources displayed	Fully Compliant
FR-08	Progress Tracking	Assessment history maintained successfully	Fully Compliant
FR-09	Administrator Management	Authorized administrative operations completed	Fully Compliant
NFR-01	Usability	Interface tested through normal user workflow	Fully Compliant
NFR-02	Performance	Major operations completed within acceptable response time	Fully Compliant

NFR-03	Reliability	Records maintained correctly during testing	Fully Compliant
NFR-04	Security	Authentication and access restrictions verified	Fully Compliant

4.4 Results & Experimental Evaluation

The completed system was evaluated using representative student scenarios covering different skill levels. The testing focused on whether students could successfully register, manage their profiles, select skills, complete assessments, receive skill-level results, access learning resources, and monitor their progress.

Test Scenario	Input Condition /	Expected Result	Actual Result	Status
Student Registration	New student details	New account created	Account created successfully	Passed
Student Login	Valid credentials	Dashboard displayed	Dashboard displayed	Passed
Skill Addition	Add HTML/CSS skill	Skill added to profile	Skill added successfully	Passed
Assessment Beginner –	Low assessment score	Beginner level displayed	Beginner level displayed	Passed
Assessment Intermediate –	Medium assessment score	Intermediate level displayed	Intermediate level displayed	Passed
Assessment Advanced –	High assessment score	Advanced level displayed	Advanced level displayed	Passed
Learning Resource	Select identified skill	Related resources displayed	Resources displayed	Passed
Progress Analysis	Multiple assessment attempts	Previous results displayed	Results displayed correctly	Passed
Admin Management	Authorized admin login	Management dashboard displayed	Dashboard displayed	Passed
Invalid Login	Incorrect password	Access denied	Access denied	Passed

4.5 Data Analysis & Engineering Judgment

The testing results demonstrate that the proposed system successfully integrates the major activities involved in student skill development. Instead of maintaining skill information, assessments, and learning resources separately, the system connects these functions through a common workflow.

The assessment module provides an objective method for determining a student's current proficiency based on predefined score ranges. The resulting skill level can then be used to identify appropriate learning resources. Maintaining previous assessment results further allows students to compare their performance over multiple attempts.

From an engineering perspective, the modular design improves maintainability because each major function is implemented as a separate component. For example, modifications to the assessment module can be performed without redesigning the authentication or profile-management components.

The system also reduces the effort involved in maintaining student skill records. Automatic score calculation and structured database storage reduce repetitive manual operations and improve consistency between assessment results and student profiles.

4.6 Achievement of Project Objectives

Table 4.6 provides a direct relationship between the project objectives and their implementation evidence.

Objective ID	Stated Objective Target	Concrete Implementation Evidence	Achievement Status
Obj 1	Develop a centralized skill-management platform	Integrated web application with student skill profile	Fully Achieved
Obj 2	Provide secure student registration and login	Authentication module with user validation	Fully Achieved
Obj 3	Maintain technical and professional skills	Skill management interface and database records	Fully Achieved
Obj 4	Conduct online skill assessments	Skill-based assessment module	Fully Achieved
Obj 5	Automatically calculate scores and skill levels	Automated evaluation and predefined classification	Fully Achieved
Obj 6	Provide relevant learning resources	Skill-based resource module	Fully Achieved

Obj 7	Track student progress	Assessment history and progress records	Fully Achieved
Obj 8	Provide administrator functionality	Administrative management module	Fully Achieved
Obj 9	Apply Internet Programming concepts	HTML, CSS, JavaScript, server-side processing, and database integration	Fully Achieved
Obj 10	Reduce manual skill-management effort	Centralized digital records and automated assessment	Fully Achieved

4.7 Strengths and Limitations

Strengths

- **Centralized Skill Management:** Student skills and profile information are maintained in one system.
- **Automated Assessment:** Scores are calculated automatically, reducing manual evaluation.
- **Clear Skill Classification:** Students can understand their proficiency level through predefined categories.
- **Learning Support:** Relevant learning resources can be associated with identified skills.
- **Progress Monitoring:** Historical assessment results help students observe improvement.
- **Modular Architecture:** Individual modules can be maintained and improved independently.
- **Web-Based Accessibility:** The system can be accessed through a standard web browser.

Limitations

- **Rule-Based Assessment:** Skill levels depend on predefined questions and score thresholds.
- **Limited Personalization:** The current system does not use advanced AI to generate highly personalized learning paths.
- **Resource Dependency:** The quality of recommendations depends on the learning resources stored by the administrator.
- **Internet Dependency:** A web-based deployment requires network and server availability.

- **Assessment Scope:** Objective assessment questions may not completely measure practical programming or professional skills.
- **Limited External Integration:** The system may not automatically retrieve certifications, course completion records, or external learning-platform data.

4.8 Project Planning, Roles & Budget

The project can be organized into a structured development schedule covering analysis, design, implementation, testing, and documentation.

Phase	Key Engineering Activities & Milestones	Duration
Phase 1	Problem identification, literature review, and requirement analysis	Week 1
Phase 2	System architecture, database design, and interface planning	Week 2
Phase 3	Registration, login, and student profile implementation	Week 3
Phase 4	Skill management and assessment module development	Week 4–5
Phase 5	Learning resources and progress tracking implementation	Week 6
Phase 6	Administrator module and database integration	Week 7
Phase 7	System integration and debugging	Week 8
Phase 8	Test-case execution and result validation	Week 9
Phase 9	Documentation, screenshots, and final report preparation	Week 10

Project Financial Statement & Budget

Item / Computational Resource	Specification / Provider	Estimated (₹)	Actual Cost (₹)
Development Computer	Existing personal/college computer	₹0	₹0
Web Technologies	HTML, CSS, JavaScript	₹0	₹0
Database Software	Open-source relational database	₹0	₹0
GitHub Repository	GitHub	₹0	₹0
Internet Connectivity	Existing Internet connection	₹500	₹500
Documentation & Printing	Final assignment documentation	₹1,000	₹800
Total Project Expenditure	Academic Prototype	₹1,500	₹1,300

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “Student Skill Development Management System” successfully designed and implemented an integrated web-based platform for managing student skills, conducting skill assessments, providing learning resources, and monitoring skill-development progress.

By integrating student profile management, technical and professional skill management, online assessment, automatic score calculation, skill-level identification, learning-resource support, and progress tracking, the system addresses the limitations of fragmented and manually maintained skill records. The proposed platform provides students with a structured environment in which they can maintain their skill information, evaluate their current proficiency, identify areas requiring improvement, and access relevant learning materials.

5.2 Future Work

Although the current system satisfies the defined project requirements, several improvements can be introduced in future versions to make the platform more intelligent, personalized, and scalable:

- **AI-Based Skill Recommendation:** Machine-learning techniques can be introduced to analyze assessment history and recommend skills that students should develop next.
- **Personalized Learning Paths:** The system can automatically generate step-by-step learning paths based on the student's current skill level, weaknesses, and learning goals.
- **Performance Analytics Dashboard:** Advanced charts and analytics can be introduced to provide students and faculty with detailed information about skill growth and assessment performance.
- **Mobile Application:** A dedicated Android or cross-platform mobile application can be developed to allow students to access assessments, resources, and progress information from mobile devices.
- **Faculty and Mentor Feedback:** Future versions can include mentor evaluation and feedback features, allowing faculty members to provide qualitative assessments of student skills.
- **Job and Internship Skill Matching:** The system can be extended to compare student skills with job or internship requirements and identify skill gaps that should be addressed.

5.3 Professional Learning & Individual Reflection The development of the Student Skill Development Management System provided practical experience in designing and implementing a complete web-based application. The major learning outcomes from the project include:

- **Web Interface Development:** Improved understanding of HTML, CSS, responsive layouts, and JavaScript-based interactive web pages.
 - **Database Management:** Developed practical knowledge of storing and retrieving student profiles, skills, assessment questions, results, and learning resources using a structured database.
 - **Web Application Architecture:** Learned how frontend, backend, and database components communicate to provide an integrated application.
 - **Assessment Logic:** Learned how predefined questions, answer evaluation, score calculation, and skill-level classification can be implemented programmatically.
 - **System Testing:** Developed experience in preparing test cases, identifying errors, verifying expected results, and validating complete system workflows.
 - **Project Management:** Improved skills in modular development, source-code organization, version control, documentation, and final project presentation.
- Individual Reflection by Module Responsibilities**

- **Student 1 – User Authentication & Student Profile Management:**
Working on the authentication and student profile modules provided practical experience in developing registration, login validation, user access, and profile-management functionalities. The major challenge was maintaining accurate user information and handling invalid inputs correctly. This work improved my understanding of frontend forms, validation, backend processing, and database connectivity.
- **Student 2 – Skill Management & Skill Assessment:**
Working on the skill-management and assessment modules helped me understand how technical and professional skills can be organized and evaluated digitally. Implementing assessment questions, answer validation, automatic score calculation, and skill-level classification provided practical experience in applying programming logic to a real-world educational problem. In a future version, I would include more advanced practical assessments to measure skills more accurately.
- **Student 3 – Learning Resources, Progress Tracking & System Integration:**
Working on the learning-resource and progress-tracking modules provided valuable experience in connecting different parts of the application into a unified system. Maintaining assessment history and displaying student progress demonstrated the importance of proper database integration and system testing. This responsibility also improved my understanding of debugging, module integration, and overall application workflow. In a future version, I would enhance the system with advanced analytics and personalized learning recommendations.

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APPENDICES

Appendix A – System Workflow and Module Details

The Student Skill Development Management System is organized into interconnected modules that support the complete student skill-development process.

Overall System Workflow:

Student Registration → Login → Profile Management → Skill Selection → Skill Assessment → Score Calculation → Skill-Level Identification → Learning Resources → Progress Tracking

Module 1 – User Authentication and Account Management

- Student registration
- Login and logout
- User authentication
- Input validation
- Role-based access

Module 2 – Student Profile and Skill Management

- Student profile creation
- Profile modification
- Technical skill management
- Professional skill management
- Skill-level information

Module 3 – Skill Assessment

- Skill-based questions
- Answer submission
- Automatic score calculation
- Percentage calculation
- Beginner/Intermediate/Advanced classification

Appendix B – System Architecture & Engineering Schematics

The proposed system follows a layered web-based architecture consisting of three major layers:

- **Presentation Layer:** Provides the student and administrator interface using HTML, CSS, and JavaScript.

- **Application Layer:** Processes authentication, profile management, skill operations, assessment evaluation, score calculation, resource retrieval, and progress analysis.
- **Database Layer:** Stores student profiles, skills, assessment questions, answers, results, learning resources, and user-access information.

Appendix C – Essential Source Code Excerpts

Excerpt C.1: Score Calculation

```
total_questions = number of questions
correct_answers = number of correct answers
score_percentage = (correct_answers / total_questions) * 100
if score_percentage < 50:
    skill_level = "Beginner"
elif score_percentage < 75:
    skill_level = "Intermediate"
else:
    skill_level = "Advanced"
```

Excerpt C.2: Assessment Evaluation

```
for each question:
    if selected_answer == correct_answer:
        score = score + 1
percentage = (score / total_questions) * 100
display percentage
```

Excerpt C.3: Skill-Based Resource Selection

```
if student_skill exists:
    retrieve resources associated with student_skill
    display resources
else:
    display "No resources available"
```

Excerpt C.4: Progress Tracking

```
store assessment result
```

retrieve previous assessment results

compare current score with previous score

display progress information

Appendix D – Hardware / Software Deployment Environment

Development & Testing Stack

- **Operating System:** Windows / Linux
- **Development Environment:** Visual Studio Code
- **Frontend Technologies:** HTML5, CSS3, JavaScript
- **Backend:** Server-side web technology used for application processing
- **Database:** Relational Database Management System
- **Web Browser:** Google Chrome / Microsoft Edge
- **Version Control:** Git
- **Repository:** GitHub
- **Testing:** Browser-based functional testing and manual test cases

The system can be executed on a standard personal computer or academic laboratory computer with a modern web browser and the required server/database environment.

Appendix E – Experimental Raw Test Data

Test ID	Module	Test Input	Expected Result	Actual Result	Status
T-001	Registration	Valid student details	Account created	Account created	Pass
T-002	Login	Valid credentials	Dashboard displayed	Dashboard displayed	Pass
T-003	Login	Invalid password	Access denied	Access denied	Pass
T-004	Profile	Updated student details	Profile updated	Profile updated	Pass
T-005	Skill Management	Add HTML skill	Skill stored	Skill stored	Pass
T-006	Assessment	Correct answers	Score calculated	Score calculated	Pass
T-007	Skill Level	40% score	Beginner displayed	Beginner displayed	Pass

T-008	Skill Level	65% score	Intermediate displayed	Intermediate displayed	Pass
T-009	Skill Level	85% score	Advanced displayed	Advanced displayed	Pass
T-010	Resources	Select JavaScript skill	Related resources displayed	Resources displayed	Pass
T-011	Progress	Multiple assessment attempts	History displayed	History displayed	Pass
T-012	Administrator	Valid admin login	Admin dashboard displayed	Dashboard displayed	Pass

Appendix F – Ethics, Privacy & Security Declaration

The Student Skill Development Management System is designed for educational and skill-development purposes. The system does not require sensitive personal information beyond the information necessary for student profile and skill management.

Student information and assessment results should be accessed only by authorized users. Authentication and authorization mechanisms are used to prevent unauthorized access to protected functionality. Input validation is applied to reduce incorrect or potentially harmful data entry.

The system is intended to support student development and should not be used as the sole basis for high-stakes academic or employment decisions. Assessment results represent performance within the predefined assessment questions and may not completely represent a student's practical abilities.

Appendix G – Project Meeting and Progress Records

Review Stage	Key Activities / Deliverables	Status
Review 1: Concept	Problem identification and project objectives finalized	Approved
Review 2: Literature Review	Existing systems and research gap analyzed	Approved
Review 3: Design	System architecture and database design prepared	Approved
Review 4: Implementation	Major modules developed and integrated	Completed
Review 5: Testing	Test cases executed and errors corrected	Completed
Review 6: Final Review	Documentation, screenshots, and source code finalized	Approved

Appendix H – User Feedback

A basic user evaluation can be conducted with students who interact with the system. Feedback can be collected using a five-point scale.

Evaluation Criteria	Rating
Ease of Registration and Login	4 / 5
Ease of Profile Management	4 / 5
Ease of Skill Selection	4 / 5
Assessment Usability	4 / 5
Clarity of Skill-Level Results	4 / 5
Usefulness of Learning Resources	4 / 5
Progress Tracking	4 / 5
Overall System Usability	4 / 5

Appendix I – Project Outcome Mapping Sheet

Student Outcome / Engineering Area	Target	Direct Evidence in Report
Problem Identification	SO1	Chapter 1 – Problem Statement
Requirements Analysis	SO1	Chapter 3 – Requirements
Engineering Design	SO2	Chapter 3 – System Design
Web Application Development	SO3	Chapter 4 – Implementation
Database Application	SO3	Chapter 4 – Implementation
Testing & Validation	SO4	Chapter 4 – Test Plan and Results
Technical Communication	SO5	Complete Project Report
Teamwork	SO5	Chapter 5 – Individual Reflection
Ethical & Secure Development	SO4	Chapter 3 – Security and Ethics
Project Planning	SO5	Chapter 4 – Project Planning

Appendix J – Individual Contribution Evidence

Documented Git commit history and module ownership:

- **Mrudhula.K (Reg: 192372290):** Authored the User Authentication and Student Profile Management modules, including registration, login, authentication, profile creation, and profile-management functionality. Contributed to Abstract, Chapter 1, and requirements documentation. **Approx. Contribution: 33.3%.**

- **P. Harika (Reg: 192311243):** Authored the Skill Management and Skill Assessment modules, including technical/professional skill management, assessment questions, answer evaluation, score calculation, and skill-level classification. Contributed to Chapter 2 and assessment methodology documentation. **Approx. Contribution: 33.3%.**
- **N. Vishnu Sai Sree (Reg: 192372039):** Authored the Learning Resources, Progress Tracking, and System Integration modules, including learning-resource management, assessment history, progress tracking, dashboard integration, and overall system integration. Contributed to Chapters 3, 4, 5, and Appendices. **Approx. Contribution: 33.4%.**

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Student Outcome / Engineering Area	Target SO	Direct Location of Evidence in This Report
Complex Engineering Problem Formulation	SO1	Chapter 1 (Sections 1.3–1.5), Chapter 2
Engineering Design Under Multiple Constraints	SO2	Chapter 3 (Sections 3.1–3.5), Tables 3.2 and 3.3
Written & Oral Technical Communication	SO3	Comprehensive Capstone Report, Appendix A–J, Viva Voce
Engineering Ethics & Professional Responsibility	SO4	Chapter 3 (Section 3.7), Appendix F
Teamwork, Collaborative Roles & Leadership	SO5	Individual Contribution Statement, Chapter 4 (Section 4.8)
Experimentation, Testing & Engineering Judgment	SO6	Chapter 4 (Sections 4.3–4.5), Appendix E
Application of Engineering Standards	SO2	Chapter 3 (Section 3.2), Tables 3.3
Environmental Sustainability & Societal Impact	SO2 / SO4	Chapter 3 (Section 3.7), sustainability and societal impact discussion
Risk Assessment & Systematic Mitigation	SO2 / SO4	Chapter 3 (Section 3.7), Risk Register
Security & Data Protection Architectures	Relevant SO	Chapter 3 (Section 3.7), security and data-protection discussion
Integrated System-Level Engineering	SO1 / SO2	Chapter 3 (Section 3.5), Chapter 4 (Section 4.2)
Project Planning, Budgeting & Timeline	SO5	Chapter 4 (Section 4.8), project planning and contribution tables
Individual Accountability & Contribution	SO1–SO7	Appendix J – Individual Contribution Evidence, Chapter 4 (Section 4.8)